Topic Choice

We were given the choice to pick whatever direction we wanted to go on based on only one criteria, all the project requirements can be implemented in it. My first idea for the project was to making a zoning program. This was inspired by a city building game. There are four zoning categories, residential, commercial, office and industrial. And each category has many sub variations. All the sub variation however have the same attribute however are different entities themselves.

Part 1

Initially this was the basis on what I started the project with simply because I already knew the bulk of the project would be the creation of said zoning categories and all aspects of the project would revolve around said portion. The four zoning categories have one super class each. All these super classes were set to abstract. The reasoning for this is that it wouldn't make sense to make an instance of the super class as you will be zoning a particular type of zoning sub variation. The super class was more of a blueprint that all the sub variations inherited its attributes and behavior from.

Residential super class would be use by Single Family Residential, Row Home, Low Rent, Medium Density and High Density sub classes.

Commercial super class would be used by Low Density Commercial and High Density Commercial sub classes.

Office super class would be used by Low Density Office, Medium Density Office, High Density Office sub classes.

Industrial super class would be used by Manufacturing Industrial, R&D Industrial and Warehouse Industrial.

Now each of these would be place in a package. So that would complete the requirement of 4 packages and 9 minimum classes. This process was aided if not possible because of the class package diagram. It was essential in visualizing the structure of the project.

For enums I have decided to make three different ones, each being an attribute. They are energy rating, safety rating and operating time.

Residential super class would have three attributes: no of families, garden space and energy rating.

Commercial super class would have three attributes: no of employees, operating time and energy rating

Office super class would have two attributes: no of employees and energy rating

Industrial super class would have three attributes: no of employees, safety rating and energy rating.

In addition, each sub class will also have two similar attributes in zoning type and zoning class. Class will refer to which category the class belongs to (Residential, Commercial, Office and Industrial). While type refers to the exact variation, or the name of the sub class (warehouse, row home and so on).

This was the bulk of the project. Once I had made all the classes in their appropriate package, I initialized each in the main class. I made an instance of each class and filled it with sample data for ease of understanding during the demo.

Part 2

The second phase of the project was to go back through the brief and implement the remaining requirements. The first one I decided to do was association because it was the “lengthiest”. For association I decided it would be based on the zoning of a street in different areas of a city/county. Suburb, outer city and inner city.

Suburb zoning keeping with the name will only have zoning for on class, single family residential. This would make it a 1 to many relationship.

Outer city zoning will have row homes, low rent homes, low density commercial and warehouse industrial zoning. This would be a 1 to many relationship to each class.

Inner city zoning would be for high density commercial, high density residential, medium density residential and high density office. This too would have a 1 to many relationship.

While programing the suburb zoning class since there was only one class type the array to hold the number of zoned house in the street was a single family residential array

(new singleFamilyResidential[] array= new singleFamilyResidential[noResidential];).

This is important as for the other two zoning classes this would not be possibe as an array cannot hold multiple different class types. This is where I will admit I stumbled across the solution. If each sub class was implemented to an interface, the interface could be used to make an array (outerCity[] ocz1=new outerCity[4]). This outer city and inner city needed to use an interface for me to be able to show their association.

Now to also fulfil the briefs requirement outer city interface has a method noiseLevel(). So each sub class with an association to outerCity zoning now has a noiseLevel() method.

Going through the rest of the requirements:

Abstract Classes were implemented into the project when each of the four super classes for the zoning categories were set to abstract because making an instance of them wouldn't make sense.

Abstract Method was implemented in the commercial super class. Each of the three sub classes overrode the abstract method in the super class with a values unique to itself in getTax().

Final Class was also implemented in the two sub classes to the commercial super class. This was because there wouldn't be any sub classes to either low or high density commercial. This could be implemented to the other subclasses but upon clarification with the lecturer it was confirmed only one instance and its appropriate reasoning was required for each of these.

Final method was implemented for offices and commercial. In commercial it was implemented in the super class so both sub classes had the same calculation for the method subsidies. The office however had the final method in each of its sub classes as they and different and unique methods.

Exception was used in the initialization on the suburb zoning and outer city zoning class. This was because there were the only two classes in which user input was asked and the bases of my implementation of exceptions were to catch specific errors in user inputs. For suburb zoning it would be entering a negative number when asked to enter number of houses in this street zoning. For outer city is would be entering a number other that 1 or 2 when asked to pick between wanting to create a sample or custom street zoning in an outer city area.

Static variable was used in single family homes to find out how many single family homes were zoned throughout the running of the main class.

Visibility of all attributes in the super class were set to protected.

Constructor chaining was used in industrial classes.